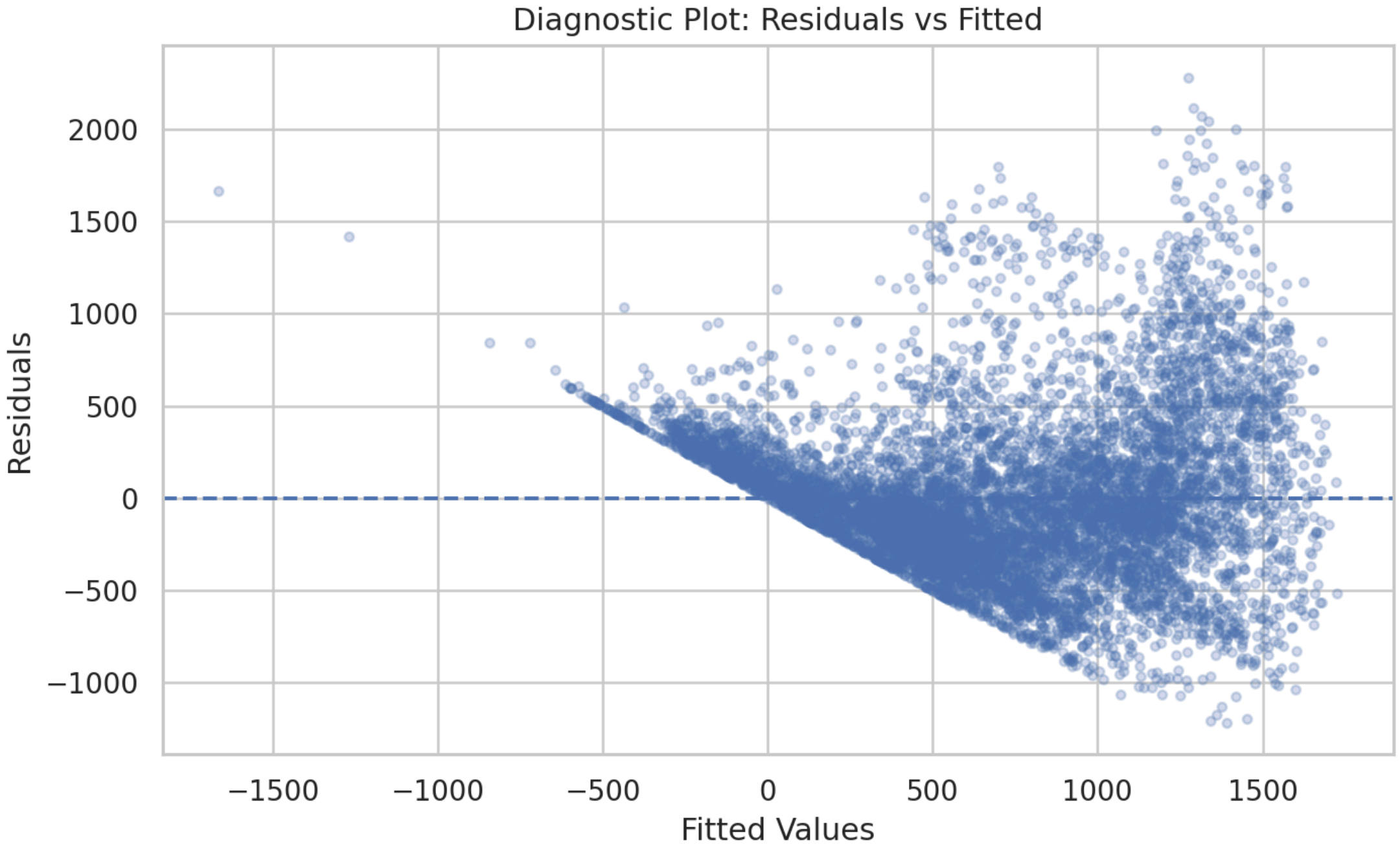
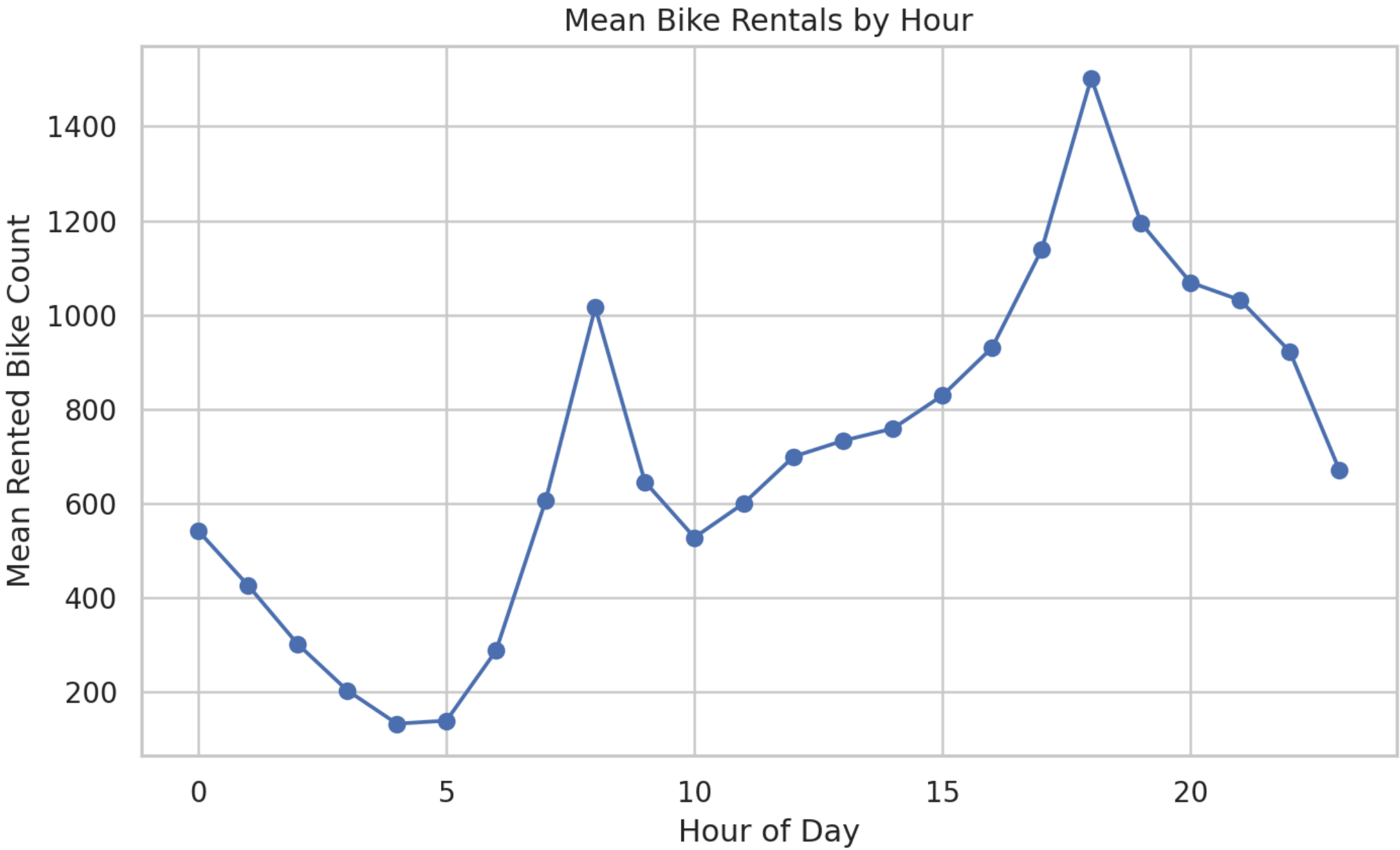
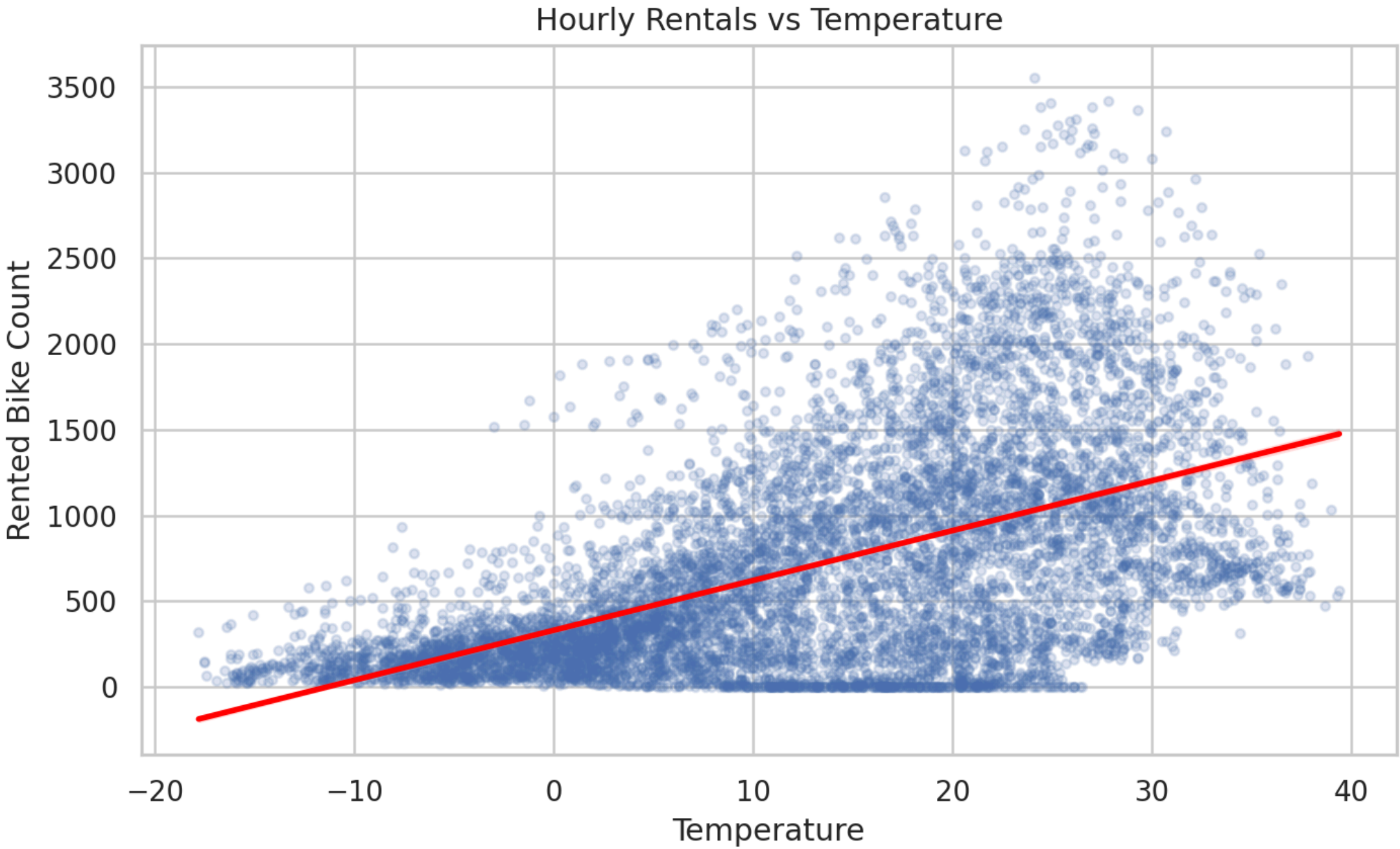
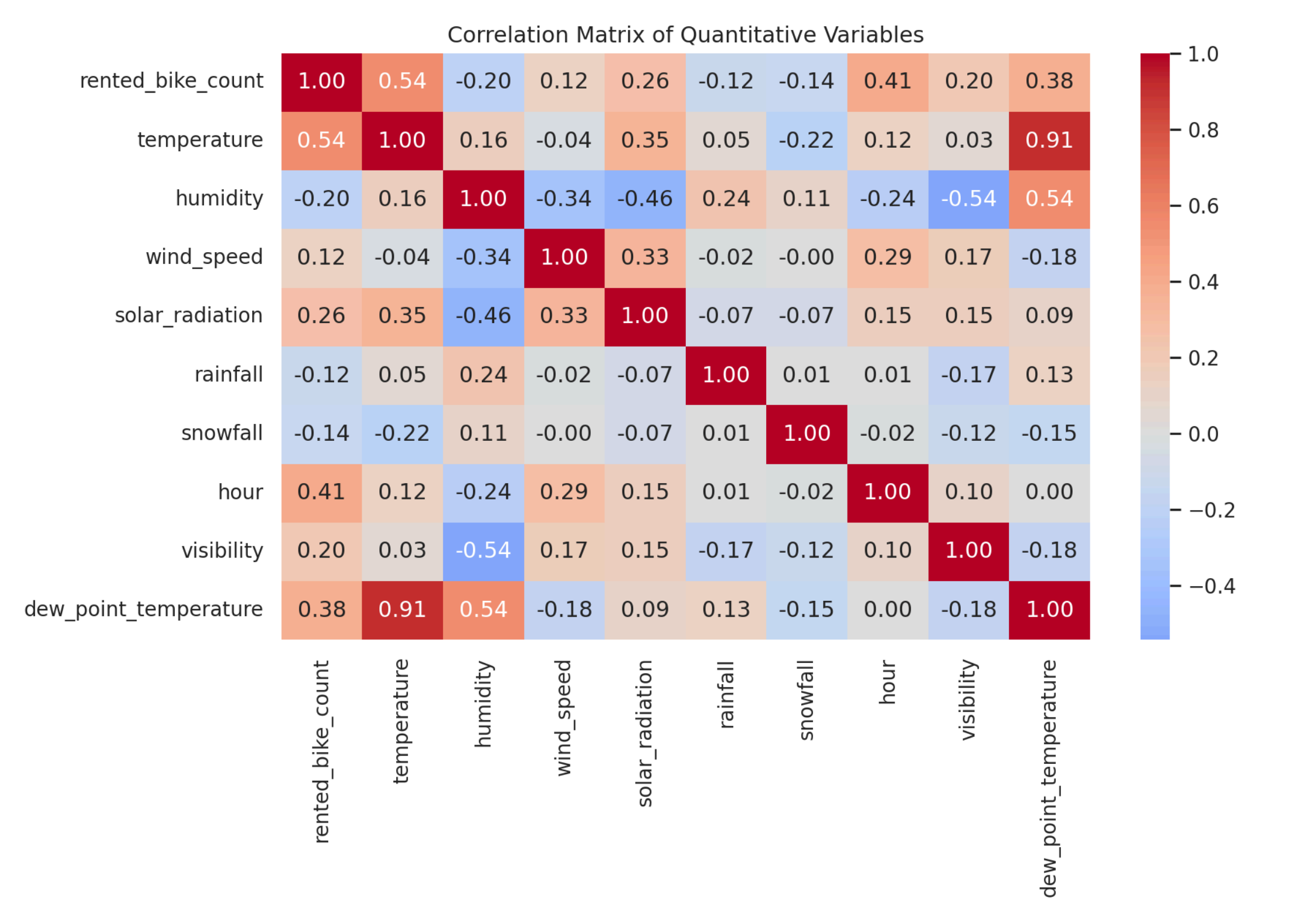
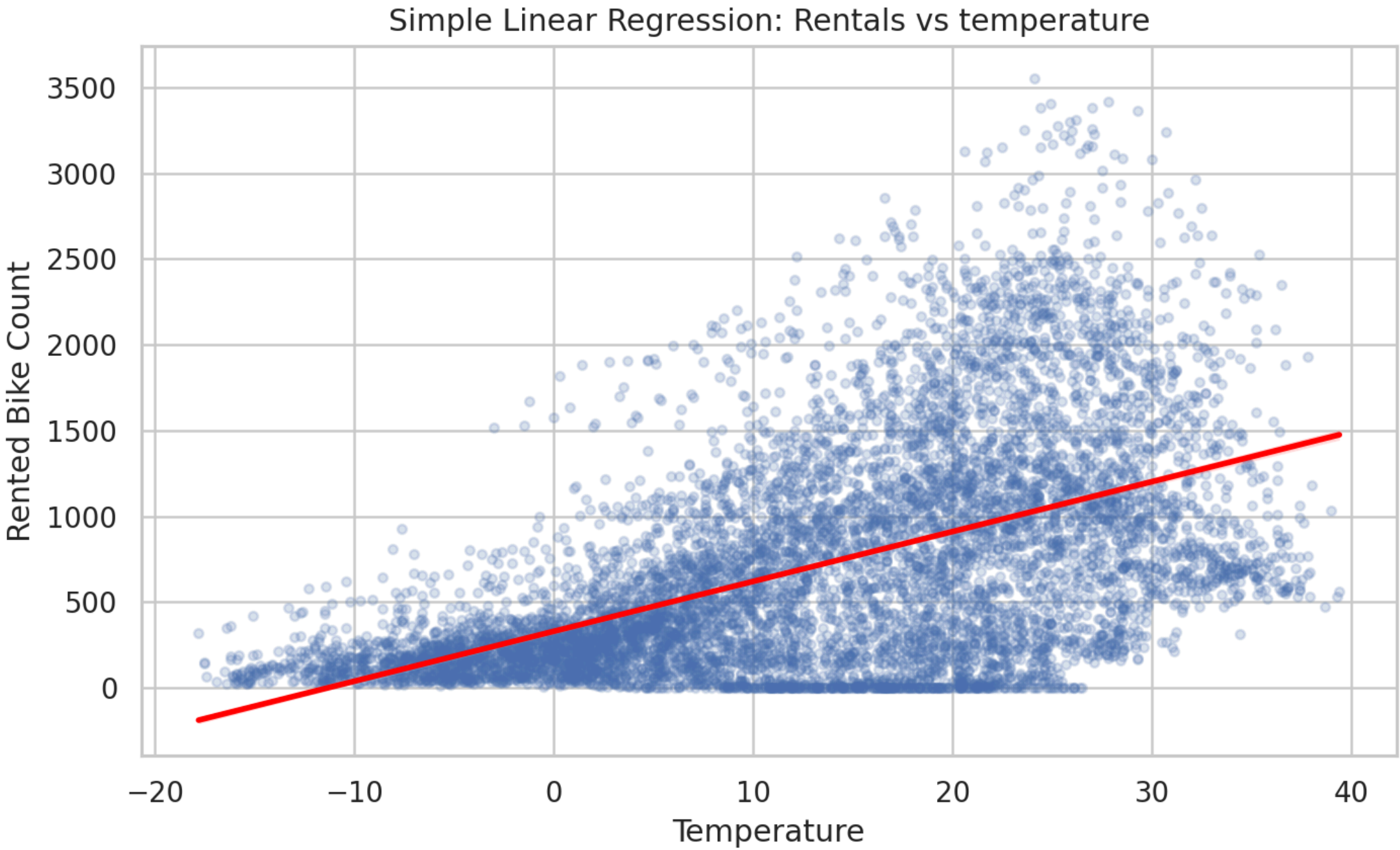
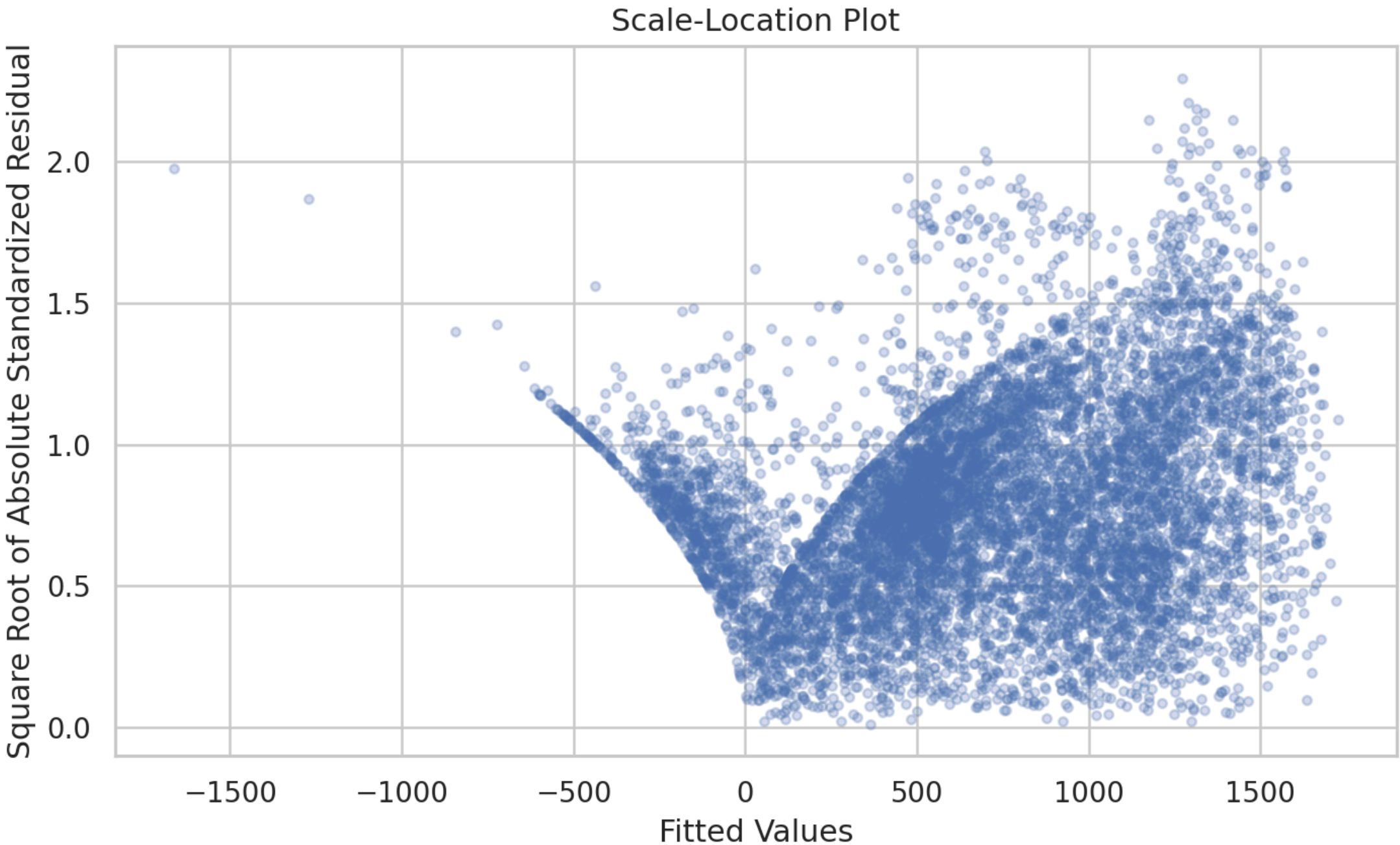

Diagnostic Plot: Residuals vs Fitted
Residuals
2000
1500
1000
500
0
−500
−1000
−1500
−1000
−500
0
500
1000
1500
Fitted Values

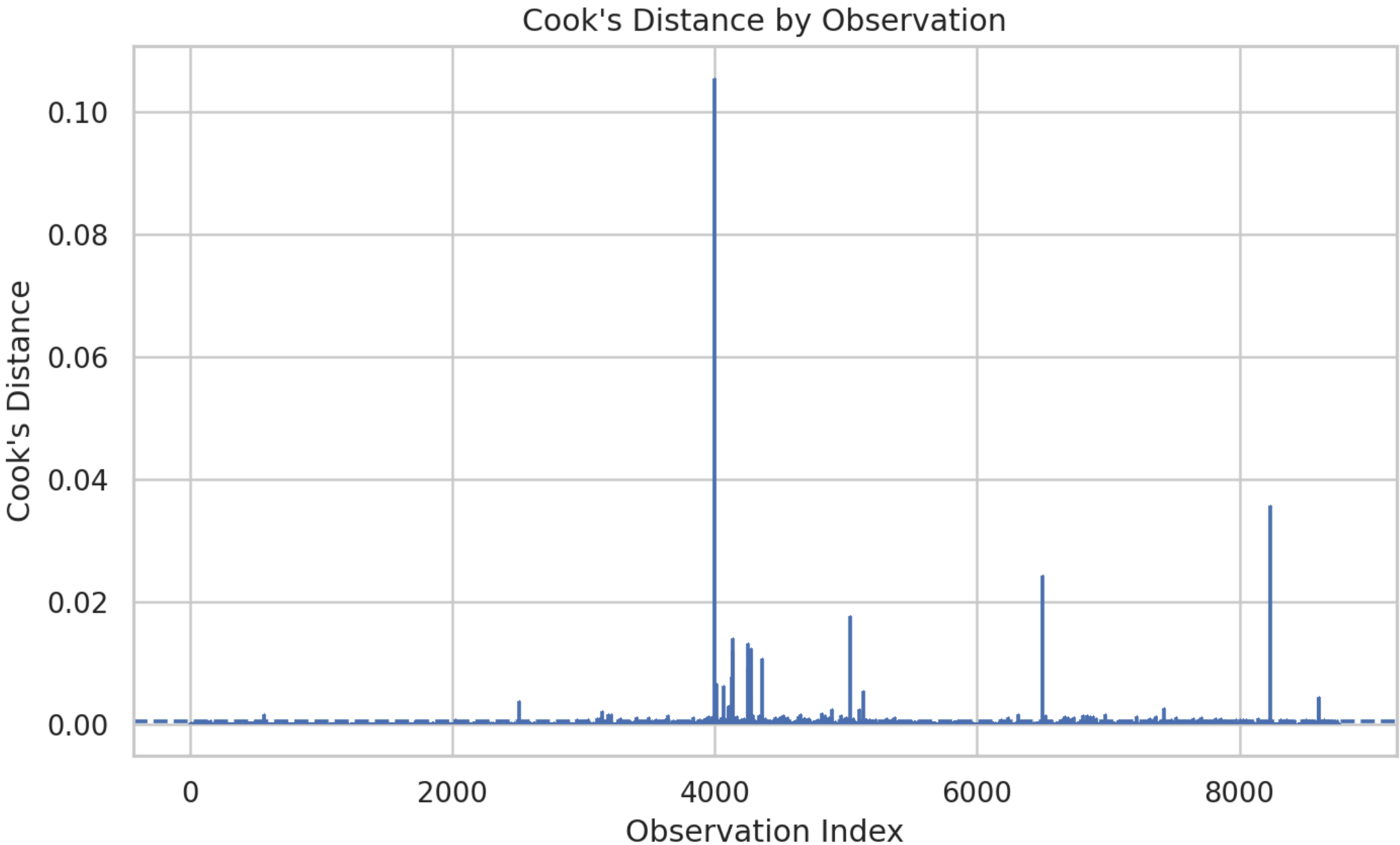

Mean Bike Rentals by Hour
Mean Rented Bike Count
1400
1200
1000
800
600
400
200
0
5
10
15
20
Hour of Day

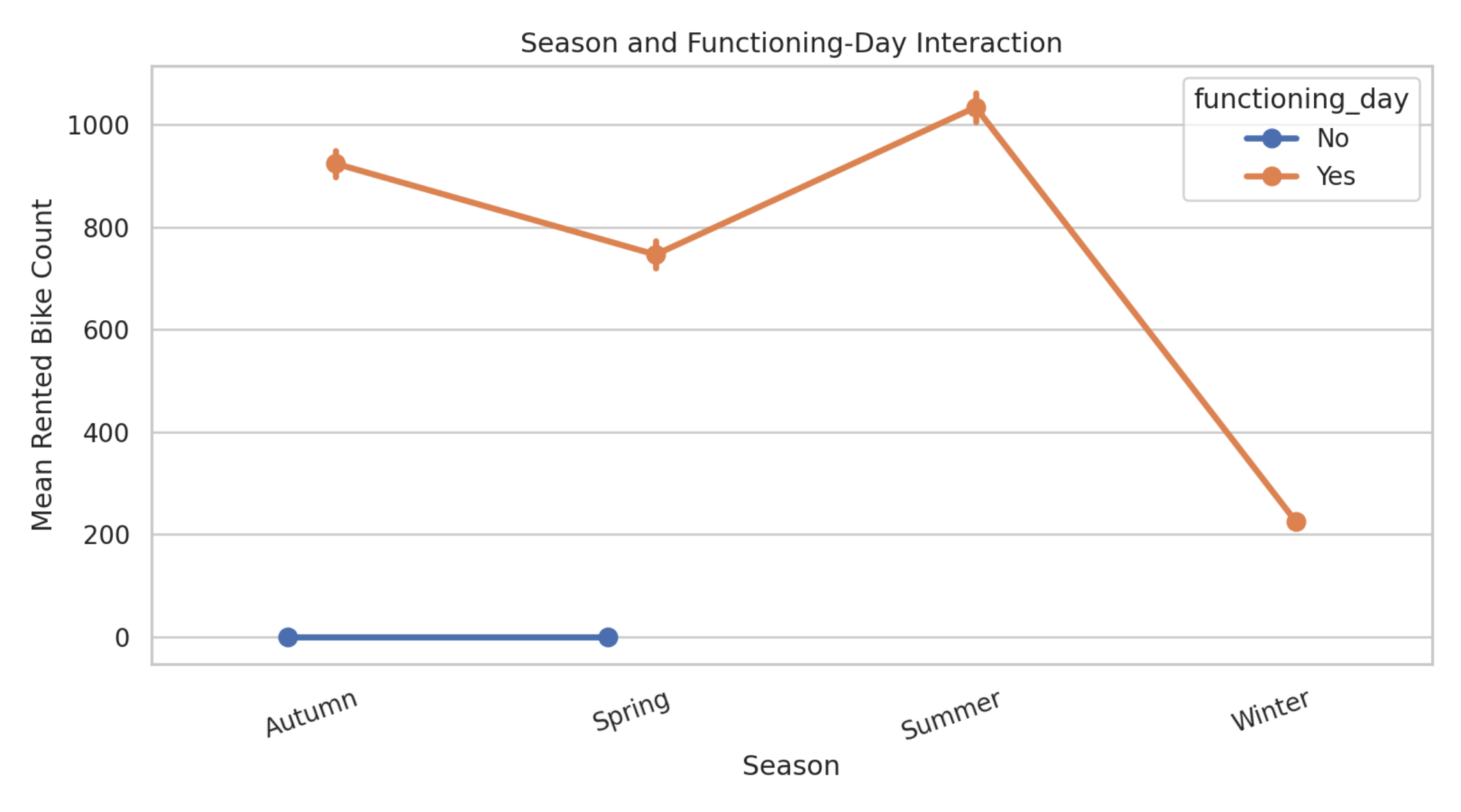

Hourly Rentals vs Temperature
Rented Bike Count
3500
3000
2500
2000
1500
1000
500
0
−20
−10
0
10
20
30
40
Temperature

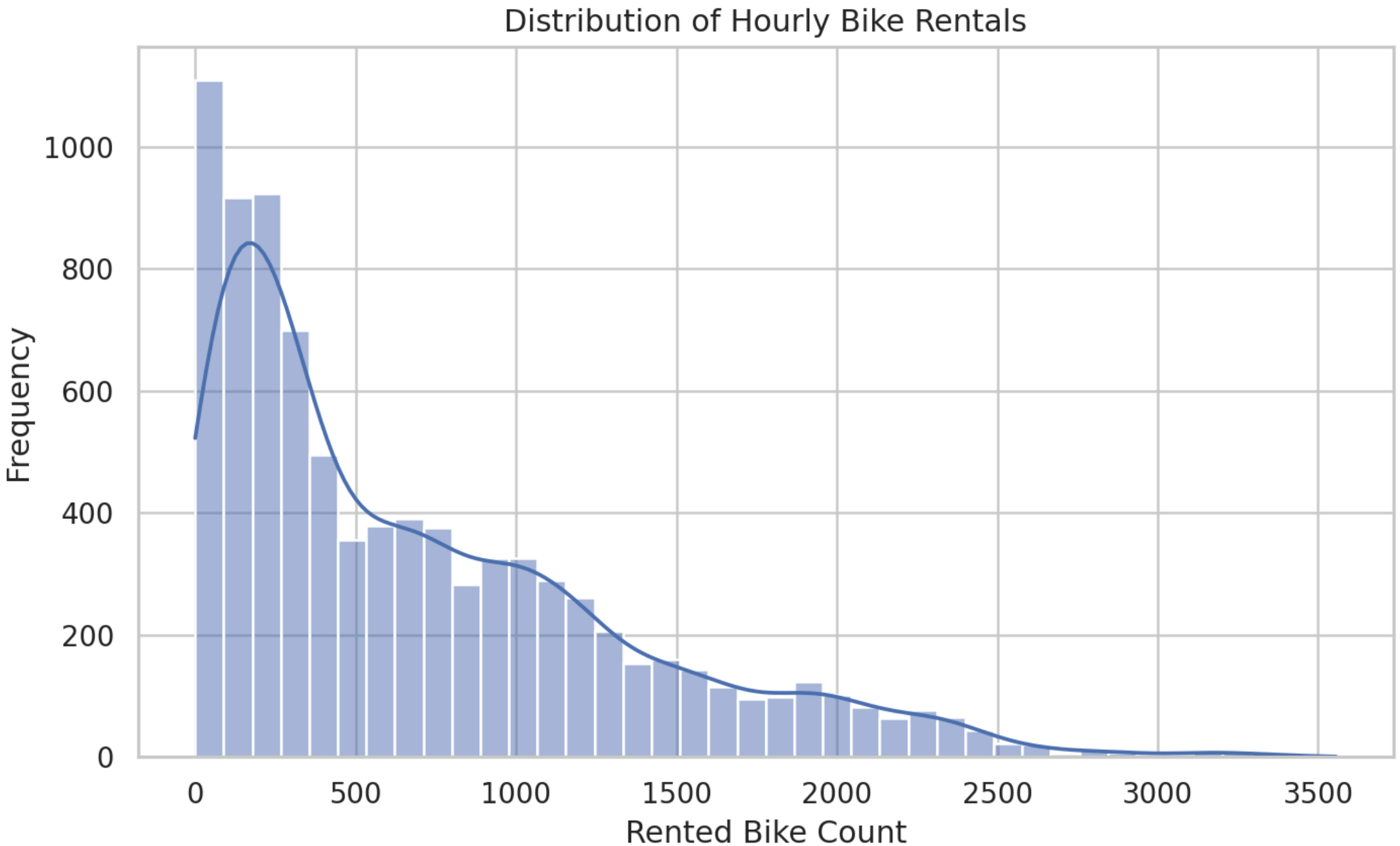

Correlation Matrix of Quantitative Variables
rented_bike_count 1.00 0.54 -0.20 0.12 0.26 -0.12 -0.14 0.41 0.20 0.38
temperature 0.54 1.00 0.16 -0.04 0.35 0.05 -0.22 0.12 0.03 0.91
humidity -0.20 0.16 1.00 -0.34 -0.46 0.24 0.11 -0.24 -0.54 0.54
wind_speed 0.12 -0.04 -0.34 1.00 0.33 -0.02 -0.00 0.29 0.17 -0.18
solar_radiation 0.26 0.35 -0.46 0.33 1.00 -0.07 -0.07 0.15 0.15 0.09
rainfall -0.12 0.05 0.24 -0.02 -0.07 1.00 0.01 0.01 -0.17 0.13
snowfall -0.14 -0.22 0.11 -0.00 -0.07 0.01 1.00 -0.02 -0.12 -0.15
hour 0.41 0.12 -0.24 0.29 0.15 0.01 -0.02 1.00 0.10 0.00
visibility 0.20 0.03 -0.54 0.17 0.15 -0.17 -0.12 0.10 1.00 -0.18
dew_point_temperature 0.38 0.91 0.54 -0.18 0.09 0.13 -0.15 0.00 -0.18 1.00
rented_bike_count
temperature
humidity
wind_speed
solar_radiation
rainfall
snowfall
hour
visibility
dew_point_temperature
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

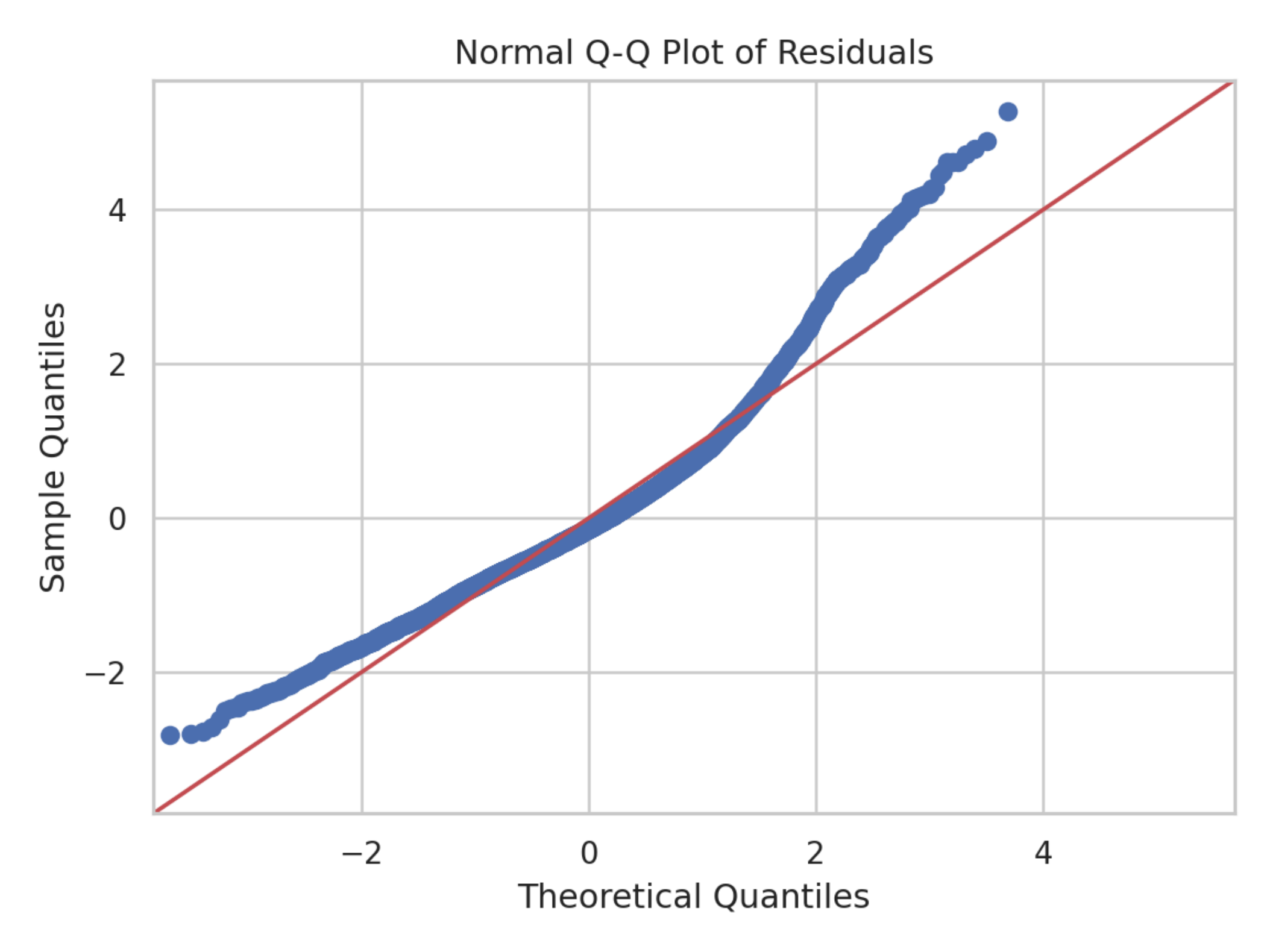

Simple Linear Regression: Rentals vs temperature
Rented Bike Count
3500
3000
2500
2000
1500
1000
500
0
−20
−10
0
10
20
30
40
Temperature

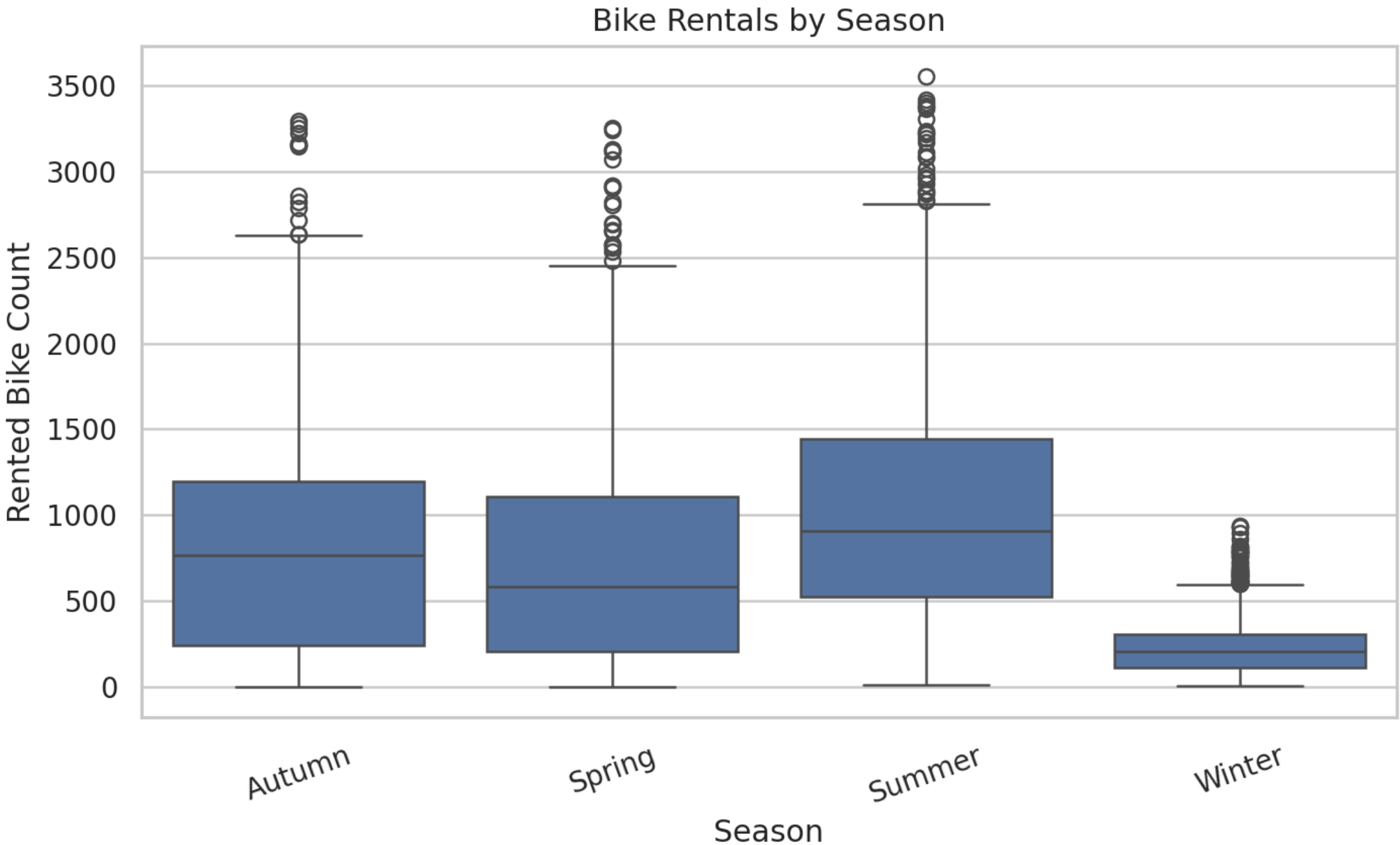

Scale-Location Plot
Square Root of Absolute Standardized Residual
2.0
1.5
1.0
0.5
0.0
−1500
−1000
−500
0
500
1000
1500
Fitted Values


Cook's Distance by Observation
Cook's Distance
0.10
0.08
0.06
0.04
0.02
0.00
0
2000
4000
6000
8000
Observation Index


Season and Functioning-Day Interaction
functioning_day
No
Yes
Mean Rented Bike Count
1000
800
600
400
200
0
Autumn
Spring
Summer
Winter
Season


Distribution of Hourly Bike Rentals
Frequency
0
200
400
600
800
1000
0
500
1000
1500
2000
2500
3000
3500
Rented Bike Count


Normal Q-Q Plot of Residuals
Sample Quantiles
Theoretical Quantiles
4
2
0
−2
−2
0
2
4


Bike Rentals by Season
Rented Bike Count
3500
3000
2500
2000
1500
1000
500
0
Autumn
Spring
Summer
Winter
Season